

PROCEDURES WITH LOWER EXPECTED SUBSET SIZE THAN THAT OF GUPTA'S SUBSET SELECTION PROCEDURE FOR THREE NORMAL POPULATIONS

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Summary: Subset selection procedures are discussed having
expected subset size uniformly below that of the
well-known procedure of Gupta for selecting the
population with the largest mean among three
normal populations.

1. INTRODUCTION

Consider three populations with distributions $N(\mu_1, \sigma^2)$, $N(\mu_2, \sigma^2)$ and $N(\mu_3, \sigma^2)$ respectively. Assume σ^2 known but $\underline{\mu} = (\mu_1, \mu_2, \mu_3)$ unknown.. The "best" population is the one having largest mean. Let X_1, X_2, X_3 be the means of samples of common size n from each of the populations. The goal is to select a subset of the three populations containing the best one with prescribed probability P^* .

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Throughout this paper we abbreviate the probability of correct selection of a given procedure by PCS and its expected subset size by ESS (of course these depend on $\underline{\mu}$, but we will not show it in our notation). A classical approach is to try to find a procedure among those satisfying the condition

$$PCS \geq P^*$$

which has a small ESS.

Gupta and Miescke (1981) show that if attention is restricted to Seal's class of procedures (see their paper for definitions) then the well-known means-procedure of Gupta (1956, 1965) has a lower ESS than any other procedure in Seal's class over much of the parameter space. The question arises: what happens when competitors outside of Seal's class is allowed? The purpose of this note is to exhibit a procedure which meets the P^* -requirement while having an ESS uniformly lower than that of Gupta's means-procedure (except when $\mu_1 = \mu_2 = \mu_3$ where equality holds). Thus, Gupta's procedure cannot be optimal in terms of ESS in any class which includes the exhibited procedure.

The means-procedure of Gupta selects all those populations whose sample means are within distance $d\sigma/n^{1/2}$ from $\max(X_1, X_2, X_3)$ with d satisfying

$$(1) \int_{-\infty}^{\infty} \Phi(x+d)^2 \phi(x) dx = P^*$$

where ϕ and Φ are the $N(0,1)$ -density and distribution function respectively.

The new procedure is essentially a modification of the multiple range (MR) procedure of Somerville (1984) and will be referred to as the MMR-procedure. To describe the procedure, let $X_{(1)} \leq X_{(2)} \leq X_{(3)}$ be the ordered sample means. Then

(a) if $X_{(3)} - X_{(1)} < d\sigma/n^{1/2}$ all three populations are selected

(b) otherwise, if $X_{(3)} - X_{(2)} < d_1\sigma/n^{1/2}$
or if $X_{(3)} - \frac{1}{2}(X_{(1)} + X_{(2)}) < d_2\sigma/n^{1/2}$

then the two populations corresponding to $X_{(2)}$ and $X_{(3)}$ are selected

(c) otherwise, only the population corresponding to $X_{(3)}$ is selected.

Here d is Gupta's constant defined by (1), $d_1 = \max(0, \sqrt{2\Phi_1(P^*)})$ and $d_2 (> d)$ is a constant chosen to make the PCS of the procedure equal to P^* when $\mu_1 = \mu_2 = \mu_3$. We give more details about d_2 in §2 below. Note that $d_1 < d$.

In §2 we also show that the PCS of this procedure is always between P^* and the PCS of Gupta's procedure and in §3 prove that the ESS of this procedure is uniformly below that of Gupta's procedure. In §4 we discuss the results.

Throughout the paper we denote by M the subset of $\{1, 2, 3\}$ corresponding to the indices of populations selected by the MMR-procedure; G denotes the corresponding subset for Gupta's procedure.

2. THE PCS OF THE MMR-PROCEDURE

In the rest of this paper we assume that $\sigma^2/n = 1$ without loss of generality. Also, for the rest of this paragraph we assume that μ_3 is the largest μ_i so that a correct selection (CS) corresponds to the event $\{3 \in M\}$ for the MMR-procedure.

We can split $\{3 \in M\}$ up into 8 subevents. Suppose that $X_2 > X_1$. Then, if $X_2 - X_1 < d$ population 3 will be selected iff $X_3 > X_2 + d$ (for, if $X_3 > X_2$ then X_3 is the largest and 3 will be selected; while if $X_2 - d < X_3 < X_2$ then it will be selected under part (a) of the Rules and if $X_3 \leq X_2 - d$ then it will be excluded under part (a)). Reasoning similarly, we find that if $X_2 > X_1$ then $\{3 \in M\}$ is the union of the following four mutually exclusive events.

$$(2) \quad E_1 = (0 < X_2 - X_1 < d_2, X_3 > X_2 - d_1)$$

$$E_2 = (d_1 < X_2 - X_1 < d_2, X_3 > X_1)$$

$$E_3 = (d_2 < X_2 - X_1 < 2d_2 - d_1, X_3 > 2X_2 - X_1 - 2d_2)$$

$$E_4 = (2d_2 - d_1 < X_2 - X_1, X_3 > X_2 - d_1)$$

Define E_i' by interchanging X_1 and X_2 in E_i for $i=1,2,3,4$. Then (apart from sets with zero probability) we have

$$(3) \quad (3 \in M) = E_1 \cup \dots \cup E_4 \cup E_1' \cup \dots \cup E_4'$$

Next introduce

$$(4) \quad Y_1 = X_1 - X_3, \quad Y_2 = X_2 - X_3$$

Then E_i and E_i' can be written in terms of Y_1 and Y_2 and $(3 \in M)$ can be represented in (Y_1, Y_2) space as the region to the lower left-hand side of the solid curve in Figure 1. Notice that it is symmetric with respect to Y_1 and Y_2 .

The event of CS for Gupta's procedure amounts to $(Y_1 < d_1, Y_2 < d)$ which is the quarter-plane to the lower left-hand side of the point (d, d) in Figure 1.

The joint distribution of (Y_1, Y_2) is bivariate normal with means $(\mu_1 - \mu_3, \mu_2 - \mu_3)$, variances 2 and covariance 1. When $\mu_1 = \mu_2 = \mu_3$ this is centred at the origin in Figure 1. Since the PCS of Gupta's procedure is P^* in this case we can get the PCS of the MMR-procedure equal to P^* in this case if we choose d_2 such that the probability contents of regions I and II in Figure 1 are equal when the joint distribution of (Y_1, Y_2) is centred at the origin. The following is a short table of values calculated by numerical integration in this way.

TABLE 1. Values of d_2

P^*	0,5	0,6	0,7	0,8	0,9	0,95	0,99	0,999
d_2	1,0186	1,3284	1,6616	2,0539	2,6026	3,0598	3,9278	4,9171

Theorem 1: The PCS of the MMR-procedure is at least P^* for any $\underline{\mu}$.

Proof: It is clear from (2) that the probability of each of the events E_i and E_i^c decreases when μ_3 decreases. Therefore it suffices to deal with the case when μ_3 equals the second largest μ_i - assume this to be μ_2 for definiteness. Then the distribution of (Y_1, Y_2) is centred at $(\nu_1, 0)$ with $\nu_1 = \mu_1 - \mu_3 \leq 0$.

Consider first the case $P^* \geq 0,5$. By the definition of d_1 the probability content below the line AB in Figure 2 is P^* for any ν_1 . It therefore suffices to show that

$$(5) \quad \int_a f(\nu_1, 0) dy_1 dy_2 \geq \int_b f(\nu_1, 0) dy_1 dy_2$$

where $f(\nu_1, \nu_2)$ denotes the joint density of (Y_1, Y_2) with $\nu_1 = \mu_1 - \mu_3$, $\nu_2 = \mu_2 - \mu_3$ ($= 0$ presently) and a and b are as indicated in Figure 2.

We have

$$(6) \quad f(\nu_1, 0)(y_1, y_2) = f(0, 0)(y_1, y_2) \exp\left(-\frac{1}{3} \nu_1^2 - \frac{1}{3} \nu_1(y_2 - 2y_1)\right).$$

Since

$$(7) \quad \int_a f(0, 0) dy_1 dy_2 = \int_b f(0, 0) dy_1 dy_2$$

by the choice of d_2 and since $\exp\left(-\frac{1}{3} \nu_1^2 - \frac{1}{3} \nu_1(y_2 - 2y_1)\right)$ is minimized over a and maximized over b at C in Figure 2 (5) follows.

For the case $P^* < 0,5$ the line AB in Figure 2 is drawn below the Y_1 -axis such that it still has P^* probability below it. Then the rest of the argument above remains true.

Theorem 2: The PCS of the MMR-procedure is at most that of Gupta's procedure for any $\underline{\mu}$.

Proof: We need to show that

$$(8) \quad \int_{I \cup III} f(\nu_1, \nu_2) dy_1 dy_2 \geq \int_{II \cup IV} f(\nu_1, \nu_2) dy_1 dy_2$$

with the regions I-IV as shown in Figure 1 and the joint distribution of (Y_1, Y_2) centred at an arbitrary point (ν_1, ν_2) in the lower left-hand quadrant. Since by symmetry

$$(9) \quad \int_{III} f(\nu_1, \nu_2) dy_1 dy_2 = \int_I f(\nu_2, \nu_1) dy_1 dy_2$$

and

$$(10) \quad \int_{IV} f(\nu_1, \nu_2) dy_1 dy_2 = \int_{II} f(\nu_2, \nu_1) dy_1 dy_2$$

we need to show

$$(11) \quad \int_I [f(\nu_1, \nu_2) + f(\nu_2, \nu_1)] dy_1 dy_2 \geq \int_{II} [f(\nu_1, \nu_2) + f(\nu_2, \nu_1)] dy_1 dy_2$$

Some algebra shows that

$$(12) \quad f(\nu_1, \nu_2)(y_1, y_2) + f(\nu_2, \nu_1)(y_1, y_2) = C f(0, 0)(y_1, y_2) g(y_1, y_2)$$

with C independent of (y_1, y_2) and

$$(13) \quad g(y_1, y_2) = \exp\left[\frac{1}{6}(y_1 + y_2)(\nu_1 + \nu_2)\right] \cosh\left[\frac{1}{2}(y_1 - y_2)(\nu_1 - \nu_2)\right].$$

Evidently g is minimized over I and maximized over II at the point D and since

$$(14) \quad \int_I f(0, 0) dy_1 dy_2 = \int_{II} f(0, 0) dy_1 dy_2$$

by the choice of d_2 , (11) follows and the theorem is proved.

It is clear from Figure 1 that the maximal difference between the PCS's of the two procedures occurs when (ν_1, ν_2) is such that the probability of I (or III) is maximal; this happens when $\nu_2 = 0$ and $\nu_1 \rightarrow -\infty$ (or $\nu_1 = 0$ and $\nu_2 \rightarrow -\infty$) i.e. when the largest two μ_i 's are equal and the smallest is infinitely smaller; then the difference is $P(d_1 < Y_2 < d) = \Phi(d/\sqrt{2}) - \Phi(d_1/\sqrt{2})$.

3. EXPECTED SUBSET SIZE

Theorem 3: The ESS of the MMR-procedure is at most that of Gupta's means-procedure for any $\underline{\mu}$.

Proof: $P(3 \in G) - P(3 \in M)$ is given by the difference between the left- and right-hand sides of (11). By interchanging the roles of the μ_i 's we obtain similar expressions for $P(i \in G) - P(i \in M)$ for $i = 1, 2$. Thus to prove the theorem we need to show that

$$(15) \quad \int_I f \, dy_1 dy_2 \geq \int_{II} f \, dy_1 dy_2$$

where

$$(16) \quad f = f(\mu_1 - \mu_3, \mu_2 - \mu_3) + f(\mu_2 - \mu_3, \mu_1 - \mu_3) + f(\mu_1 - \mu_2, \mu_3 - \mu_2) \\ + f(\mu_3 - \mu_2, \mu_1 - \mu_2) + f(\mu_2 - \mu_1, \mu_3 - \mu_1) + f(\mu_3 - \mu_1, \mu_2 - \mu_1).$$

It involves no loss of generality to assume that

$$(17) \quad \mu_1 + \mu_2 + \mu_3 = 0$$

since only the differences of the μ_i 's are involved. (16) can be written as

$$(18) \quad f(y_1, y_2) = C f_{(0,0)}(y_1, y_2) h(y_1, y_2)$$

where

$$(19) \quad h(y_1, y_2) = \exp(\mu_1 y_1 + \mu_2 y_2) + \exp(\mu_1 y_2 + \mu_2 y_1) \\ + \exp(\mu_1 y_1 + \mu_3 y_2) + \exp(\mu_1 y_2 + \mu_3 y_1) \\ + \exp(\mu_2 y_1 + \mu_3 y_2) + \exp(\mu_2 y_2 + \mu_3 y_1).$$

Using the identity $\exp(A) + \exp(B) = 2\exp(\frac{1}{2}(A+B))\cosh(\frac{1}{2}(A-B))$ with

different choices of pairs, h can be written in various forms, among which

$$(20) \quad h(y_1, y_2) = 2\exp(-\frac{1}{2}\mu_3(y_1+y_2))\cosh(\frac{1}{2}(y_1-y_2)(\mu_1-\mu_2)) \\ + 2\exp(-\frac{1}{2}\mu_2(y_1+y_2))\cosh(\frac{1}{2}(y_1-y_2)(\mu_1-\mu_3)) \\ + 2\exp(-\frac{1}{2}\mu_1(y_1+y_2))\cosh(\frac{1}{2}(y_1-y_2)(\mu_2-\mu_3))$$

and

$$(21) \quad h(y_1, y_2) = 2\exp(\mu_3(y_2 - \frac{1}{2}y_1))\cosh(\frac{1}{2}y_1(\mu_1-\mu_2)) \\ + 2\exp(\mu_2(y_2 - \frac{1}{2}y_1))\cosh(\frac{1}{2}y_1(\mu_1-\mu_3)) \\ + 2\exp(\mu_1(y_2 - \frac{1}{2}y_1))\cosh(\frac{1}{2}y_1(\mu_2-\mu_3)).$$

(21) shows that along lines $y_2 = \frac{1}{2}y_1 + k$ h increases as y_1 decreases from zero to more negative values. Thus, in Figure 1 the maximum value of h over region II is on its lower boundary (DF) and the minimum over I is on its upper boundary (ED). (20) shows that h increases if (y_1, y_2) is moved perpendicularly away from the line $y_1 = y_2$. If (z_1, z_2) is on DF, let (z'_1, z'_2) be the projection perpendicular to $y_1 = y_2$ of (z_1, z_2) onto DH. Then $h(z_1, z_2) \leq h(z'_1, z'_2) \leq h(z''_1, z''_2)$ with (z''_1, z''_2) the co-ordinates of D. Hence h is maximized over II at D. Similarly, h is minimized over I at D.

Using (14) and (18), (15) and therefore the theorem follows.

4. DISCUSSION

- Intuitively, if two of the three means are fairly close together and both much larger than the third, then a "reasonable" procedure should treat the problem essentially as one in which selection among only two populations is involved. Gupta's procedure does not have this feature while the MMR has it: when $X_{(1)}$ is very much smaller than $X_{(2)}$ and $X_{(3)}$ Gupta's procedure selects only the largest when $X_{(3)} - X_{(2)} > d\sigma/\sqrt{n}$

whereas the MMR-procedure does this when $X_{(3)} - X_{(2)} > d_1 \sigma / n^{\frac{1}{2}}$. This observation partly explains the possibility of finding a procedure that dominates Gupta's procedure in terms of expected subset size. It is only a partial explanation, however, since shrinking certain regions requires expanding other regions to maintain the required PCS and it is somewhat surprising that this can be done without increasing the ESS anywhere.

2. Suppose we replace d_1 in the definition of the MMR-procedure by an arbitrary d_1^* with $d_1 \leq d_1^* \leq d$ and correspondingly replace d_2 by d_2^* chosen such that the P^* -requirement is met at $\mu_1 = \mu_2 = \mu_3$. Then $d \leq d_2^* \leq d_2$ and we have a class of procedures indexed by d_1^* having MMR and Gupta's procedures as extreme members (the cases $d_1^* = d_1$ and $d_1^* = d$ respectively). Then it is not difficult to show both the PCS and the ESS of any member of this class is between those of MMR and Gupta's procedure. The arguments required are similar to those given above. Hence MMR is the best procedure within this class in the sense that while meeting the P^* -requirement it has the lowest ESS uniformly in $\underline{\mu}$.
3. It is clear from the geometric nature of our proofs that other procedures can also be constructed which meet the P^* -requirement and have lower ESS compared to Gupta's procedure. We are not suggesting that MMR is a particularly recommendable procedure but chose it because it is easy to parameterise and not too difficult to compare with Gupta's procedure.

5. ACKNOWLEDGEMENT

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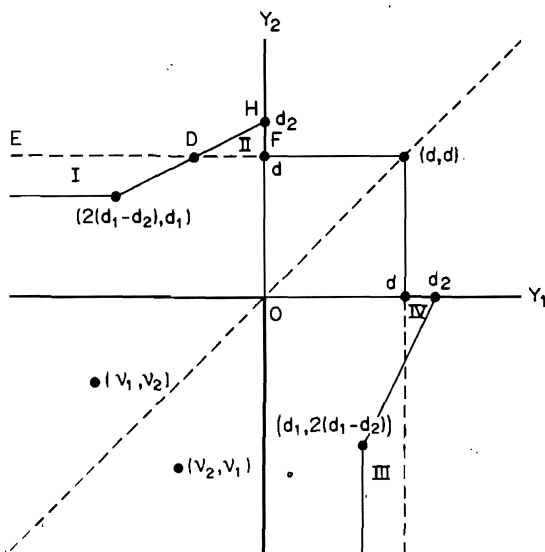


FIGURE 1. Regions of correct selection

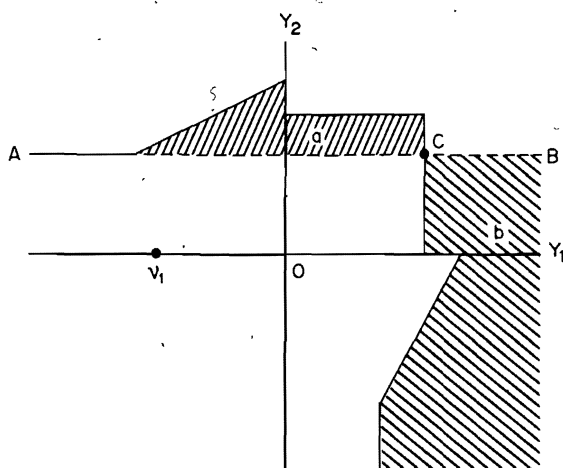


FIGURE 2. Proof that $PCS \geq P^*$